Junho Andrew Choi

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SUMMARY

Data Science and Integrative Biology senior at UC Berkeley targeting machine learning, applied AI, and AI evaluation roles. Two research internships building and stress-testing production AI systems: large language model evaluation on a live enterprise system, and deep learning for 3D medical imaging, with a habit of measuring whether a model is actually right rather than trusting that it passed.

EDUCATION

University of California, Berkeley Expected May 2027
B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology *Berkeley, CA*
Coursework: Machine Learning (CS 189, in progress), Data Engineering (DS C101, in progress), Principles & Techniques of Data Science, Data Structures, Probability for Data Science, Optimization Modeling

SKILLS

Languages: Python, SQL, Java, R
Machine Learning & AI: PyTorch, scikit-learn, NumPy, pandas, deep learning, large language model (LLM) evaluation, retrieval-augmented generation (RAG), natural language to SQL (Text2SQL), knowledge graphs (Neo4j), model interpretability (SHAP, Grad-CAM), feature engineering
Data & Analytics: PostgreSQL, exploratory data analysis (EDA), statistics, data pipelines, model evaluation, data visualization, Jupyter, Git, React, Vite, Figma
Spoken: Korean (native), English (fluent), Mandarin

EXPERIENCE

SmartMind AI May 2026 - Aug 2026
AI Researcher, Data Science Team *Seoul, Republic of Korea*
- Initiated and led a natural-language-to-SQL (NL2SQL) pipeline on a live enterprise ERP system, replacing single-run pass/fail testing with a 43-run reproducibility framework (15 scenarios, 3 trials each) that produced a measured 13/15 pass rate and exposed 3 accounting-logic defects producing valid but silently wrong SQL
- Improved retrieval quality in a Neo4j ontology pipeline (ApeRAG/Qurify) by auditing entity-merge accuracy and tuning similarity thresholds, tracing failed queries to missing or duplicate nodes, and migrating 15 business rules from prompts into structured graph nodes

Asan Medical Center (AMC) May 2025 - Aug 2025
AI Research Intern *Seoul, Republic of Korea*
- Co-authored a research paper on a multimodal deep survival model for hepatocellular carcinoma, developing and fine-tuning 3D U-Net and ResNet3D architectures in PyTorch and integrating SHAP and Grad-CAM to validate feature importance for high-stakes clinical prediction
- Ran large-scale exploratory analysis across 3D CT and clinical modalities for the VLM3D Multi-Abnormality Classification Challenge, informing the feature selection and strategy that placed 7th among multi-institutional teams

Invention Corps of Berkeley - Mozilla Firefox Aug 2025 - Feb 2026
Design Engineer Consultant *Berkeley, CA*
- Engineered a Firefox performance-monitoring extension in React and Vite that tracked, filtered, and sorted 50+ concurrent tabs by live memory and CPU usage
- Led survey-based user research with a 12-person team, translating findings into feature requirements and Figma interface designs that shaped the shipped extension

PROJECTS

- **rad-DINO:** implemented and ran experiments with a self-supervised vision transformer for radiology imaging, evaluating learned representations for downstream classification in PyTorch
- **Multi-Agent Ad Generation System:** built an 8-agent large language model pipeline for on-brand ads, where agents exchange a versioned JSON spec rather than prose and a brand-guardian agent hard-vetoes on deterministic color and layout gates
- **AI Tactical Assistant:** trained a PyTorch network to predict match outcomes from historical player and team data, paired with a PuLP linear-programming optimizer generating constrained optimal formations
- **Activation Domain Prediction Pipeline:** developed a CNN-BiLSTM model classifying transcriptional activation domains from protein sequences with sliding-window feature engineering and K-means clustering, reaching F1 = 0.95 across 10 subclasses

LEADERSHIP & AWARDS

US Army, 8th Army 2ID HHBN Sep 2022 - Mar 2024
Non-Commissioned Officer *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean-English platoon leader, delivering Army Combat Fitness Test (ACFT) training and coursework to a 96% platoon pass rate
- Awarded the US Army Certificate of Achievement (Mar 2024) for leadership and bilingual communication